

PrepBuddy

Pakistan's Premier MDCAT & NUMS Study Platform

Full Platform Proposal & Development Blueprint

Prepared for Antigravity Development Team

Version 1.0 • April 2026

4

40+

FastAPI + Next.js

Groq LLM

Development Phases

Platform Features

Tech Stack

AI Engine

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1. Executive Summary

PrepBuddy is a comprehensive, AI-powered EdTech platform purpose-built for Pakistani students preparing for MDCAT (Medical and Dental College Admission Test) and NUMS (National University of Medical Sciences) entrance examinations. Every year, over 200,000 students compete for a limited number of medical college seats in Pakistan. The preparation journey is hampered by scattered resources, lack of organized past papers, absence of personalised feedback, and no structured self-assessment tools.

PrepBuddy solves all of this in one beautifully designed, student-friendly platform. It combines a rich past-paper bank, a subject-wise and chapter-wise quiz engine, AI-generated explanations powered by Groq LLM, gamification (streaks, leaderboards, friend competitions), real-time performance analytics, and a news feed — delivering everything a student needs to crack MDCAT or NUMS in one place.

■ Core Purpose	MDCAT & NUMS Preparation Platform for Pakistani Students
■ AI Engine	Groq LLM (primary) — scalable to GPT in later phases
■ Tech Stack	FastAPI (Backend) + Next.js 14 (Frontend) + PostgreSQL
■ Platforms	Web (desktop + mobile responsive) — PWA ready
■ Market	200,000+ MDCAT aspirants annually in Pakistan
■ Timeline	24 Weeks across 4 Development Phases
■ Auth	JWT-based Auth with Google OAuth option
■ Analytics	Real-time performance dashboard with AI insights

2. Problem Statement

1. Scattered Past Papers

Students spend hours searching across WhatsApp groups, Telegram channels, and random websites for past papers. There is no single, verified, well-organised repository for MDCAT/NUMS past papers.

2. No Personalised Practice

Generic YouTube videos and PDF books do not adapt to a student's weak areas. There is no tool that says: 'You are weak in Genetics — here are targeted MCQs.'

3. Zero Progress Tracking

Students have no way to measure their improvement over time. They do not know their accuracy per subject, speed per question, or readiness level before the exam.

4. No Motivation Mechanism

Preparation for 1–2 years is mentally exhausting. Without streaks, leaderboards, or friend competitions, students lose focus and motivation easily.

5. Lack of Instant Explanations

When a student gets an MCQ wrong, they often have no immediate, clear explanation of the correct answer with scientific reasoning — forcing them to search elsewhere.

6. No Real-Time MDCAT News

Policy changes, schedule updates, and seat allocation news are critical — yet there is no single trusted place where students get MDCAT/NUMS news alongside their study tools.

3. Solution Overview – PrepBuddy

PrepBuddy is a single, unified platform that addresses every pain point above. Here is how each problem maps to a PrepBuddy feature:

Feature	Description
Scattered Past Papers	Organised Past Paper Bank (2020–2024+), filterable by year, subject, chapter
No Personalised Practice	AI-Powered Quiz Engine with weak-area detection and Groq LLM explanations
Zero Progress Tracking	Performance Dashboard with accuracy, speed, subject-wise breakdown
No Motivation Mechanism	Daily Streaks, XP Points, Leaderboards, Friend Challenges
No Instant Explanations	Groq LLM generates instant answer explanations for every MCQ
No MDCAT News	Dedicated MDCAT/NUMS News Feed section with latest updates

Beyond solving existing problems, PrepBuddy also introduces features that students do not even know they need yet — such as AI-generated mock tests, study time heatmaps, avatar customisation, and friend competition rooms — making the platform addictive in a productive way.

4. Target Audience & Market

Feature	Description
Primary Users	MDCAT aspirants — FSc Part I & II students (ages 16–22) across Pakistan
Secondary Users	NUMS aspirants and students repeating entry tests
Tertiary Users	Academy teachers and tutors who want to assign quizzes to students
Admin Users	PrepBuddy content team who upload and manage past papers and MCQs
Market Size	200,000+ MDCAT registrations per year; growing at ~8% annually
Geography	All provinces of Pakistan, with highest concentration in Punjab, KPK, Sindh
Device Usage	~70% mobile users, ~30% desktop — platform must be fully responsive
Monetisation	Freemium model: Basic access free, Premium subscription for full features

5. Complete Feature Set

5.1 Landing / Marketing Website

The first impression of PrepBuddy must be powerful enough to convert a visitor into a registered user. The landing page is a fully animated, high-conversion marketing website.

Feature	Description
Hero Section with Animation	Animated headline: 'Crack MDCAT 2025 — Pakistan's Smartest Prep Platform'. Animated counter showing total questions, students, and papers available.
Feature Showcase	Visual cards showing: Past Papers, AI Quiz, Streaks, Friend Competition, Analytics.
How It Works Section	3-step animated flow: Register → Choose Subject → Start Practicing.
Statistics Bar	Live counters: Questions attempted today, students active, papers available.
Testimonials / Social Proof	Student testimonial cards with avatars and score improvements.
MDCAT Countdown Timer	Live countdown to the next MDCAT exam date — creates urgency and keeps students returning.
Subject Preview	Interactive peek at Biology, Chemistry, Physics, English with sample MCQs — no login required.
Call-to-Action (CTA)	Prominent 'Start Free' button. Sticky top navbar with Login/Register buttons.
SEO Optimised	Full Next.js SSR/SSG for search engine optimisation. Meta tags for MDCAT, NUMS keywords.
Dark Mode Toggle	Landing page supports light/dark mode for visual comfort.

5.2 Student Authentication & Onboarding

Feature	Description
Email Registration	Standard email + password signup with email verification OTP.
Google OAuth	One-click signup / login via Google account.
Phone OTP Login	Pakistani phone number OTP login (via Twilio or local gateway).

Feature	Description
Onboarding Wizard	5-step wizard after first login: Avatar, Target Exam, Subjects, Study Goal, Daily Target.
Profile Picture Upload	Student uploads a photo or picks from 20+ built-in avatars.
Password Reset	Secure forgot-password flow via email link.
Remember Me	JWT with refresh tokens — stay logged in across sessions.
Session Management	View and revoke active sessions from profile settings.

5.3 Student Profile & Dashboard

Feature	Description
Profile Card	Avatar, name, target exam, joining date, city, XP level badge.
Quick Stats Widget	Today's questions attempted, current streak, total XP, rank.
Subject Progress Rings	Circular progress rings for Biology, Chemistry, Physics, English — showing % mastery.
Recent Activity Feed	Chronological log of quizzes taken, papers viewed, achievements unlocked.
Achievement Badges	Earned badges displayed on profile: 'Week Warrior', 'Biology Master', '1000 MCQs', etc.
Study Plan Section	Personalised weekly study plan based on weak areas and exam date.
Goal Tracker	Daily MCQ target progress bar — e.g. '47 / 100 MCQs done today'.
Friend List on Profile	Show number of friends and mini-avatars of top 5 friends.
Public Profile Option	Students can make their profile public for friend invitations.
Profile Edit	Update avatar, bio, city, target exam date, daily MCQ goal.

5.4 Past Paper Bank

The Past Paper Bank is the core content repository of PrepBuddy. It solves the single biggest complaint of MDCAT students — scattered, unverified past papers.

Feature	Description
Year-wise Papers	MDCAT & NUMS past papers from 2015 to present, organised by year.
Subject Filter	Filter papers by: Biology, Chemistry, Physics, English, or All Subjects.
Chapter Filter	Drill down to a specific chapter within a subject — e.g. Cell Biology.
PDF Viewer (In-Platform)	Embedded PDF viewer — students read papers inside the app, no download required.
Download Option	Premium students can download PDFs for offline use.
Paper Bookmarking	Bookmark specific papers to revisit later.
Question Highlight Tool	Students highlight specific questions in the PDF viewer for review.
Difficulty Tag	Each paper tagged with difficulty level: Easy / Medium / Hard.
Answer Key Overlay	Toggle to show/hide answer key alongside the paper.
Paper Rating	Students rate the difficulty of a paper (1–5 stars) — crowdsourced data.
Search within Papers	Full-text search across all past papers to find specific topics.
Admin Upload Panel	Admin uploads PDFs with metadata: Year, Exam, Subjects covered, Chapter tags.
Paper Views Counter	Show how many students have viewed/attempted each paper.
Related Papers	After viewing a paper, suggest related papers from the same subject/year.

5.5 Quiz & MCQ Engine

The Quiz Engine is the heart of PrepBuddy. It delivers personalised, subject-wise and chapter-wise MCQs with instant AI-powered explanations.

Feature	Description
Subject Selection	Choose one or multiple subjects: Biology, Chemistry, Physics, English.
Chapter Selection	Select specific chapter(s) within a subject — e.g. Enzymes in Biology.
Topic Selection	Further narrow down to a specific topic — e.g. 'Competitive Inhibition'.
Question Count Setting	Set number of questions: 10, 20, 30, 50, or Custom.
Timed Mode	Optional countdown timer per question (30s / 60s / 90s) or full quiz timer.
Untimed Practice Mode	Relaxed mode with no timer — for learning new topics.
4-Option MCQ Display	Clean, tap-friendly MCQ UI with A/B/C/D options — large touch targets.
Instant Feedback	After selecting an answer: green = correct, red = wrong, correct option shown.
AI Explanation (Groq)	Every question has an AI-generated explanation via Groq LLM — shown after answer.
Question Flagging	Flag a question if it seems incorrect — sent to admin for review.
Question Bookmarking	Bookmark individual MCQs to a personal revision list.
Quiz Result Summary	End screen: Score, Time taken, Correct/Wrong/Skipped, Subject breakdown.
Retry Wrong Questions	One-click button to retry only the questions answered incorrectly.
Quiz History	Full log of past quizzes with scores, dates, and subjects.
Adaptive Difficulty	System serves harder questions on topics where student scores >80%, easier on <50%.
Custom Question Mix	Student can mix subjects — e.g. 30 Biology + 20 Chemistry in one session.
Leaderboard per Quiz	After a quiz, see how your score compares to friends who took the same quiz.

5.6 Mock Test Mode

Feature	Description
Full MDCAT Mock	Complete 200-question mock test simulating the actual MDCAT format and timing.
NUMS Mock	NUMS-specific mock with 100 questions and NUMS subject weightings.
Subject-Wise Mock	30-question mini-mock for one subject only — perfect for targeted practice.
Timed Countdown	Exact MDCAT timing: 200 questions in 150 minutes (45 seconds/question average).
No Going Back Mode	Strict exam mode: cannot change answer once selected — trains exam discipline.
Review Mode After Submit	After submission, full review of all questions with explanations.
Score Prediction	AI predicts likely MDCAT score range based on mock performance.
Percentile Rank	Show student's percentile rank among all PrepBuddy students who took the mock.
Mock Scheduling	Set a recurring weekly mock test — platform sends reminder notification.
Mock History	All past mocks stored with scores, timing, and subject performance.
Weak Area Report	After each mock, AI generates a personalised weak area report.
Print Result	Students can print or download their mock test result as PDF.

5.7 AI-Powered Features (Groq LLM)

Groq LLM provides near-instant AI inference, making it ideal for real-time explanations and insights. PrepBuddy uses Groq as the primary AI engine with a clear upgrade path to GPT-4 for advanced reasoning tasks in later phases.

Feature	Description
Instant MCQ Explanations	After every question, Groq generates a 2–4 sentence explanation of the correct answer with scientific reasoning.
AI Study Coach	Chat interface where students ask questions: 'Explain the difference between mitosis and meiosis.'

Feature	Description
Weak Area Detection	After 20+ questions, AI identifies and highlights weak chapters with a ranked list.
Personalised Study Plan	AI generates a week-by-week study plan based on exam date, weak areas, and daily time available.
Mnemonics Generator	Student selects a topic → Groq generates a memorable mnemonic — e.g. for amino acid classification.
Concept Summary Cards	AI generates a one-page summary card for any chapter — useful for last-minute revision.
Difficulty Calibration	AI monitors performance and adjusts question difficulty in real-time.
Performance Narrative	Weekly AI-written summary: 'This week you improved in Organic Chemistry but need work on Genetics.'
Exam Readiness Score	AI calculates a 0–100 readiness score based on mock results and practice history.

5.8 Streak & Gamification System

Gamification is the key driver of daily retention. Students who maintain streaks are 5x more likely to continue using a platform consistently. PrepBuddy's gamification system is designed to be addictive in a healthy, motivating way.

Feature	Description
Daily Streak Counter	Fire emoji streak counter on dashboard — maintained by attempting at least 10 MCQs per day.
Streak Freeze Token	Students earn 1 freeze token per week — used to protect streak on a missed day (like Duolingo).
XP Points System	Earn XP for every: MCQ correct (+5), quiz completed (+20), mock completed (+100), daily goal hit (+30).
Level System	XP thresholds unlock levels: Beginner → Learner → Scholar → Expert → MDCAT Champion.
Achievement Badges	30+ unique badges: '7-Day Streak', 'Biology Ace', 'Speed Demon', 'First Mock', etc.
Daily Challenge	A special daily challenge of 5 tough MCQs — earns 2x XP if all correct.
Weekly Goals	Set a weekly MCQ target — progress bar on dashboard with celebration animation on completion.
Leaderboard	Global and friends-only leaderboard ranked by XP — resets weekly to keep it fair.
Rewards Shop (Phase 3)	XP points redeemable for: extra freeze tokens, premium mock tests, avatar items.
Milestone Celebrations	Confetti animation when student hits 100, 500, 1000 MCQ milestones.
Streak Share	One-tap share streak card to WhatsApp/Instagram — viral growth mechanic.

5.9 Friends & Competition System

Feature	Description
Friend Invite via Link	Unique referral link — friend clicks it, registers, and is auto-added to friend list.
WhatsApp Share	Shareable invite card optimised for WhatsApp sharing.

Feature	Description
Friend Search	Search PrepBuddy users by username or phone number to send a friend request.
Friend List	View friends' avatars, current streaks, XP, and last active time.
1v1 Quiz Challenge	Challenge a specific friend to a 20-question quiz in a chosen subject — real-time or async.
Challenge Notification	Friend receives a push notification: 'Ali challenged you! Can you beat his score of 85%?'
Challenge Leaderboard	After both complete, a comparison card shows both scores, time, accuracy.
Group Competition Room	Create a room with 2–10 friends, choose subject, all take same quiz simultaneously.
Room Leaderboard	Live leaderboard during group quiz updates as participants submit answers.
Friend Activity Feed	See friends' recent achievements and completed quizzes — social motivation.
Challenge History	Full log of all challenges: won, lost, pending — with detailed comparison.
Block/Remove Friend	Privacy controls: remove a friend, block unwanted users.

5.10 Performance Analytics Dashboard

Feature	Description
Overall Accuracy Rate	Big-number display: e.g. '74% accuracy across 2,340 MCQs attempted.'
Subject-Wise Accuracy	Bar chart comparing accuracy in Biology, Chemistry, Physics, English.
Chapter-Wise Breakdown	Drill-down table showing accuracy for every chapter within a subject.
Accuracy Over Time	Line chart showing accuracy trend — weekly average over last 12 weeks.
Speed Analytics	Average time per question — track improvement in answering speed.
Strengths & Weaknesses	AI-curated list: Top 3 strong chapters and Top 3 weak chapters.
Question Attempt Heatmap	Calendar heatmap (like GitHub) showing MCQs attempted per day.
Subject Pie Chart	Pie chart of time spent per subject — helps identify subject neglect.
Mock Score History	Line graph of all mock test scores over time.
Comparison with Average	See how your score compares to the PrepBuddy platform average.
Predicted MDCAT Score	AI prediction: 'Based on your performance, you are likely to score 150–165/200.'
Printable Report	Export full performance report as PDF — useful for academy teachers.

5.11 Study Time Tracker

Feature	Description
Auto Session Tracking	Platform auto-tracks time spent in quizzes, paper reading, and AI chat.
Daily Study Time Widget	Dashboard widget: 'You have studied for 2h 15m today.'
Weekly View	Bar chart of daily study hours for the current week.

Feature	Description
Monthly View	Calendar view of study time per day — identify productive vs weak days.
Subject Time Breakdown	Pie chart: How much time spent on each subject this week.
Study Goals	Set a daily study time goal — e.g. 3 hours/day — with progress ring.
Idle Detection	If student is inactive for 5 minutes, timer pauses — accurate tracking.
Study Streaks from Time	Streak also maintained by studying for minimum 30 minutes/day.

5.12 MDCAT & NUMS News Section

Feature	Description
News Feed	Curated list of latest MDCAT, NUMS, and PMC policy news — updated daily.
Breaking News Banner	Red banner on dashboard for critical announcements — e.g. 'MDCAT 2025 date announced.'
Category Filters	Filter news by: Schedule, Policy Changes, Result Announcements, Tips & Tricks.
Save Article	Bookmark important news articles for later reading.
Exam Date Calendar	Official calendar of upcoming MDCAT and NUMS key dates.
PMC Updates	Direct integration with PMC (Pakistan Medical Commission) announcements.
Student Blog	PrepBuddy-authored blog posts: preparation tips, strategies, toppers' stories.
Push Notifications	Breaking news notifications: Students opt-in to receive instant alerts.

5.13 Admin Dashboard

Feature	Description
Secure Admin Login	Separate admin panel at /admin — JWT-protected with role-based access.
Past Paper Upload	Upload PDFs with metadata: Year, Exam Type, Subjects, Chapters, Difficulty.
MCQ Management	Add/Edit/Delete individual MCQs with: Question, 4 Options, Correct Answer, Explanation, Subject, Chapter, Topic.
Bulk MCQ Import	Upload MCQs in bulk via CSV/Excel template — instant processing.
Student Management	View all registered students: filter by activity, subject, registration date.
Content Moderation	Review flagged questions, approve/reject student-submitted corrections.
Analytics Overview	Admin dashboard showing: total users, daily active users, top quizzes, popular papers.
News Management	Create, edit, schedule, publish news articles and blog posts.
Announcement Broadcast	Send platform-wide push notifications to all students.
Subject & Chapter CRUD	Manage subject taxonomy: add new subjects, chapters, topics.
User Roles	Roles: Super Admin, Content Admin, Moderator — different permission levels.
Revenue Dashboard (Ph 3)	Track premium subscriptions, trial conversions, revenue by month.
Export Reports	Download user activity and content usage reports as Excel/CSV.

5.14 Notifications & Reminders

Feature	Description
Daily Study Reminder	Push notification if student has not studied by a set time — customisable.
Streak at Risk Alert	Notification: 'Your streak is at risk! You have 3 hours left to practice.'
Friend Challenge Alert	Instant notification when a friend sends a quiz challenge.

Feature	Description
Achievement Unlock	In-app celebration + notification when a badge is earned.
Weekly Report	Every Monday: personalised weekly performance email report.
New Paper Available	Notification when admin uploads a new past paper for a subscribed subject.
Exam Countdown	Daily notification: 'X days to MDCAT — you have studied Y hours this week.'
Notification Preferences	Student controls which notifications to receive and at what times.

5.15 Accessibility, Mobile & UX

Feature	Description
Fully Responsive	Next.js responsive design — works flawlessly on mobile, tablet, and desktop.
Progressive Web App	PWA support — students can 'install' PrepBuddy on their phone home screen.
Dark Mode	Full platform dark mode toggle — easy on eyes during late-night study.
Urdu Language Option	Phase 3: Toggle between English and Urdu interface language.
Font Size Control	Students adjust text size for better reading comfort.
Offline Mode (PWA)	Phase 4: Cached quizzes available offline — useful for low-connectivity areas.
Keyboard Navigation	Full keyboard navigation for accessibility compliance.
Loading Skeletons	Skeleton screens instead of spinners — premium UX feel.
Micro-animations	Framer Motion animations on quiz transitions, streak updates, badge unlocks.
Error Handling	Friendly error pages — no raw error messages shown to students.

6. Tech Stack & Architecture

Layer	Technology	Purpose
Frontend	Next.js 14 (App Router)	React-based SSR/SSG framework — SEO-friendly, fast
Frontend	TypeScript	Type safety across all frontend code
Frontend	Tailwind CSS	Utility-first styling — fast UI development
Frontend	Framer Motion	Smooth animations for streaks, badges, transitions
Frontend	React Query / TanStack	Server state management, caching, pagination
Frontend	Recharts / Chart.js	Performance analytics charts and graphs
Backend	FastAPI (Python)	Async REST API — high performance, auto-generated docs
Backend	PostgreSQL	Primary relational database
Backend	Redis	Caching, session management, real-time leaderboards
Backend	Celery + Redis	Background tasks: email sending, report generation
AI	Groq API (Llama 3)	Ultra-fast LLM inference for MCQ explanations
Auth	JWT + OAuth2 (Google)	Secure authentication system
Storage	AWS S3 / Cloudflare R2	PDF storage and media uploads
Email	Resend / SendGrid	Transactional emails and weekly reports
Push Notif	Firebase Cloud Messaging	Mobile and web push notifications
DevOps	Docker + Docker Compose	Containerised development and deployment
DevOps	GitHub Actions	CI/CD pipeline for automated testing and deployment
Hosting	Vercel (Frontend)	Next.js deployment with edge functions

Layer	Technology	Purpose
Hosting	Railway / DigitalOcean (Backend)	FastAPI + PostgreSQL hosting
Monitoring	Sentry	Error tracking and performance monitoring

Architecture follows a decoupled microservices-ready monolith pattern: FastAPI backend exposes a clean REST API consumed by the Next.js frontend. The AI service (Groq) is called server-side to protect API keys. Redis handles real-time features. All PDFs are stored in object storage (S3/R2) and served via CDN for fast loading across Pakistan.

7. Database Schema Overview

The following are the core database tables. Each table has standard audit fields: `id` (UUID), `created_at`, `updated_at`, `is_deleted`.

users

`id`, `email`, `phone`, `password_hash`, `name`, `avatar_url`, `city`, `target_exam`, `xp_points`, `level`, `streak_count`, `streak_last_date`, `is_premium`, `role`

subjects

`id`, `name`, `slug`, `icon_url`, `color_hex`

chapters

`id`, `subject_id`, `name`, `slug`, `order_index`

topics

`id`, `chapter_id`, `name`, `slug`

questions

`id`, `subject_id`, `chapter_id`, `topic_id`, `question_text`, `option_a`, `option_b`, `option_c`, `option_d`, `correct_option`, `explanation`, `difficulty`, `year`, `source`, `is_active`

past_papers

`id`, `title`, `exam_type`, `year`, `subject_ids[]`, `pdf_url`, `answer_key_url`, `difficulty`, `view_count`, `download_count`

quizzes

`id`, `user_id`, `mode`, `subject_ids[]`, `chapter_ids[]`, `question_count`, `score`, `accuracy`, `time_taken_sec`, `completed_at`

quiz_answers

`id`, `quiz_id`, `question_id`, `selected_option`, `is_correct`, `time_taken_sec`

bookmarks

`id`, `user_id`, `question_id`, `paper_id`, `created_at`

achievements

`id`, `user_id`, `badge_slug`, `earned_at`

friends

`id`, `user_id`, `friend_id`, `status` (`pending/accepted/blocked`)

challenges

`id`, `challenger_id`, `challenged_id`, `subject_id`, `status`, `challenger_score`, `challenged_score`, `expires_at`

study_sessions

id, user_id, session_start, session_end, duration_minutes, subject_id

notifications

id, user_id, type, title, body, is_read, sent_at

news_articles

id, title, slug, body, category, thumbnail_url, published_at, author_id

streak_freezes

id, user_id, used_on_date, remaining_count

8. API Endpoints Overview

FastAPI auto-generates Swagger UI at /docs. All endpoints use /api/v1/ prefix. Protected routes require Bearer JWT token.

Auth

[POST /auth/register](#) | [POST /auth/login](#) | [POST /auth/google](#) | [POST /auth/refresh](#) | [POST /auth/forgot-password](#) | [POST /auth/reset-password](#)

User Profile

[GET /users/me](#) | [PATCH /users/me](#) | [GET /users/{id}/public](#) | [POST /users/me/avatar](#) | [GET /users/me/stats](#)

Subjects & Chapters

[GET /subjects](#) | [GET /subjects/{id}/chapters](#) | [GET /chapters/{id}/topics](#) | [GET /topics/{id}/questions](#)

Quiz Engine

[POST /quizzes/start](#) | [POST /quizzes/{id}/answer](#) | [POST /quizzes/{id}/complete](#) | [GET /quizzes/history](#) | [GET /quizzes/{id}/review](#) | [GET /quizzes/{id}/ai-explanation](#)

Mock Tests

[POST /mocks/start](#) | [POST /mocks/{id}/submit](#) | [GET /mocks/history](#) | [GET /mocks/{id}/result](#)

Past Papers

[GET /papers](#) | [GET /papers/{id}](#) | [GET /papers/{id}/download](#) | [POST /papers/{id}/bookmark](#) | [POST /papers/{id}/rate](#)

Streaks & Gamification

[GET /streaks/me](#) | [POST /streaks/freeze](#) | [GET /achievements/me](#) | [GET /leaderboard/global](#) | [GET /leaderboard/friends](#)

Friends & Challenges

[GET /friends](#) | [POST /friends/invite](#) | [PATCH /friends/{id}/accept](#) | [POST /challenges/create](#) | [GET /challenges/incoming](#) | [POST /challenges/{id}/complete](#)

Analytics

[GET /analytics/overview](#) | [GET /analytics/subjects](#) | [GET /analytics/chapters/{subject_id}](#) | [GET /analytics/heatmap](#) | [GET /analytics/study-time](#)

AI Features

[POST /ai/explain/{question_id}](#) | [POST /ai/study-plan](#) | [POST /ai/weak-areas](#) | [POST /ai/coach/chat](#) | [POST /ai/mnemonic](#) | [GET /ai/readiness-score](#)

News

[GET /news](#) | [GET /news/{slug}](#) | [POST /news/{id}/bookmark](#) | [GET /news/categories](#)

Admin

*POST /admin/papers/upload | POST /admin/questions | POST /admin/questions/bulk-import | GET /admin/users |
POST /admin/news | GET /admin/analytics/overview | POST /admin/notifications/broadcast*

9. Development Phases

PrepBuddy is developed across 4 structured phases over 24 weeks. Each phase delivers a working, deployable version of the platform with increasing feature richness.

PHASE 1 | Foundation & Core Infrastructure

Weeks 1–6 | Deliverable: Working MVP with Auth, Past Papers, Basic Quiz

Phase 1 establishes the entire technical foundation. By end of Phase 1, a student can register, view past papers, and take a basic subject-wise quiz.

✓ Project setup: FastAPI + Next.js + PostgreSQL + Docker	✓ Database schema — all core tables defined and migrated
✓ Landing page with hero, features, stats, CTA, countdown timer	✓ Student registration (email + OTP verification)
✓ Student login (email/password + Google OAuth)	✓ JWT auth with refresh token mechanism
✓ Onboarding wizard (5-step profile setup)	✓ Student profile page with avatar and basic stats
✓ Subject/Chapter/Topic management (Admin CRUD)	✓ MCQ management — Admin add/edit/delete questions
✓ Bulk MCQ import via CSV	✓ Past Paper upload by Admin (PDF + metadata)
✓ Past Paper Bank page — list, filter by year/subject	✓ Embedded PDF viewer for past papers
✓ Basic Quiz Engine — subject + chapter + question count	✓ MCQ display UI — 4 options, timer, navigation
✓ Instant answer feedback (correct/incorrect highlight)	✓ Quiz result summary screen
✓ Admin dashboard — user list, paper list, MCQ list	✓ Basic deployment: Vercel + Railway + PostgreSQL

PHASE 2 | AI, Gamification & Analytics

Weeks 7–12 | Deliverable: Full-featured platform with AI + Streaks + Analytics

Phase 2 adds the AI brain (Groq), the gamification layer, performance analytics, friend system, and mock test mode — transforming MVP into a complete product.

✓ Groq LLM integration — MCQ explanations per question	✓ AI Study Coach chat interface
✓ Weak area detection algorithm	✓ AI-generated personalised study plan

✓ Daily streak system with fire counter	✓ Streak freeze tokens (earn + spend)
✓ XP points system — all earning events	✓ Level system (Beginner → MDCAT Champion)
✓ 30+ achievement badges with unlock logic	✓ Daily challenge (5 special MCQs — 2x XP)
✓ Weekly goal system with progress bar	✓ Global + Friends leaderboard (XP-based, weekly reset)
✓ Performance analytics dashboard — all charts	✓ Subject-wise and chapter-wise accuracy breakdown
✓ Study time tracker (auto-session tracking)	✓ Activity calendar heatmap
✓ Full MDCAT Mock Test mode (200Q / 150 min)	✓ NUMS Mock Test mode
✓ Mock test result with percentile rank	✓ Adaptive difficulty algorithm
✓ Friend invite via referral link + WhatsApp share	✓ Friend list + friend activity feed
✓ 1v1 Quiz Challenge system	✓ Challenge notifications (push + in-app)
✓ Question bookmarking	✓ Quiz retry (wrong answers only)
✓ Push notification system (Firebase)	✓ Email: weekly performance report
✓ MDCAT News section — curated feed	✓ Dark mode — full platform

PHASE 3 | Advanced Features & Monetisation

Weeks 13–18 | Deliverable: Premium tier, Group Competitions, AI Enhancements

Phase 3 introduces revenue generation through the premium subscription model, advanced AI features, group competition rooms, and Urdu language support.

✓ Premium subscription system (Stripe / EasyPaisa)	✓ Freemium feature gating logic
✓ XP Rewards Shop (redeem tokens for perks)	✓ Group Competition Rooms (2–10 friends)
✓ Live leaderboard during group quiz	✓ AI Score Prediction ('You will likely score 160–170')
✓ AI Readiness Score (0–100 exam readiness)	✓ AI Mnemonic Generator per topic
✓ AI Concept Summary Cards (chapter revision)	✓ Full Urdu language interface toggle
✓ Printable performance report (PDF export)	✓ Subject-wise mini-mock tests (30 questions)
✓ Answer key overlay in PDF viewer	✓ Question highlight tool in PDF viewer
✓ Paper rating system (crowdsourced difficulty)	✓ Paper bookmarking
✓ Full-text search across past papers	✓ PMC announcements feed integration
✓ Exam Date official calendar	✓ Streak share card (WhatsApp/Instagram)
✓ Milestone celebration animations (confetti)	✓ Admin: Revenue dashboard
✓ Admin: Content moderation queue	✓ Admin: Scheduled news publishing
✓ Admin: Push notification broadcaster	✓ Academy/Teacher role (assign quizzes to students)
✓ Student report sharing with teacher	✓ Google Analytics + Mixpanel integration

PHASE 4 | Scale, Polish & Future-Proofing

Weeks 19–24 | Deliverable: PWA, Offline Mode, Video Lectures, Platform Scale

Phase 4 polishes every edge, adds offline capability, prepares for viral growth, and introduces video learning and advanced social features.

✓ Progressive Web App (PWA) — installable on phone	✓ Offline mode — cached quizzes available offline
✓ Video lecture integration (embedded YouTube/Vimeo)	✓ Video + MCQ pairing (watch lecture → take quiz on it)

✓ AI-powered video timestamp bookmarks	✓ Live Class scheduling (Zoom/Google Meet integration)
✓ Student Community Forum (subject-wise discussions)	✓ Forum upvoting, pinning, moderation
✓ GPT-4 upgrade option for complex AI queries	✓ AI-generated full mock paper (from question bank)
✓ Adaptive mock test based on weaknesses	✓ Multi-city leaderboard (Lahore vs Karachi vs Islamabad)
✓ National rank badge on profile	✓ Monthly MDCAT Grand Challenge (all PrepBuddy users)
✓ Referral rewards system (invite friends → earn XP)	✓ Parent dashboard — track child's study time
✓ SMS OTP for rural areas with poor internet	✓ NUMS-specific past paper bank expansion
✓ AKU, Agha Khan entry test content	✓ Performance benchmarking against top scorers
✓ Automated platform health monitoring (Sentry + PagerDuty)	✓ Load testing and scalability improvements
✓ GDPR-compliant data export (student data portability)	✓ Full accessibility audit (WCAG 2.1 AA)

10. Non-Functional Requirements

Feature	Description
Performance	Page load < 2 seconds on 4G. API response < 300ms for 95% of requests. Groq AI response < 1.5 seconds.
Scalability	Architecture supports 50,000 concurrent users. Horizontal scaling via Docker + load balancer. Database connection pooling.
Availability	99.9% uptime target. Separate staging and production environments. Database daily backups with 30-day retention.
Reliability	Graceful error handling — never show raw errors to students. Retry logic for AI API calls. Circuit breaker for Groq API.
SEO	Next.js SSR for all public pages. Meta tags, Open Graph, sitemap.xml, robots.txt. Core Web Vitals score > 90.
Internationalisation	Text strings externalised for Urdu translation in Phase 3. RTL support considered in layout design.
Browser Support	Latest 2 versions of Chrome, Firefox, Safari, Edge. Mobile: iOS 15+, Android 10+.

11. Security & Compliance

Feature	Description
Authentication	JWT access tokens (15-min expiry) + refresh tokens (7-day). Bcrypt password hashing (cost factor 12).
Authorisation	Role-based access control: Student, Teacher, Content Admin, Super Admin. Row-level security in DB.
API Security	Rate limiting: 100 req/min per IP. CORS whitelist. Input validation on all endpoints.
Data Protection	HTTPS enforced everywhere. Sensitive data encrypted at rest. No plaintext passwords ever stored.
PDF Security	PDFs served via signed S3 URLs (15-min expiry) — cannot be hotlinked or shared externally.

Feature	Description
Admin Security	Admin panel on separate subdomain. 2FA required for admin login. All admin actions logged.
Payment Security	Payment processing via Stripe (PCI-DSS compliant) — PrepBuddy never stores card data.
Privacy	No student data sold to third parties. Privacy policy clearly displayed. Data deletion on request.

12. Deliverables Checklist

The following deliverables are expected from the Antigravity development team at the completion of each phase and at final handover:

Feature	Description
Source Code	Full source code in a private GitHub repository with branch protection rules.
Documentation	API documentation (Swagger UI), README for local setup, deployment guide.
Database Migrations	Alembic migration files for all database changes — rollback-safe.
Admin Credentials	Initial super admin account credentials and instructions.
Seed Data	Seed script with: 5 subjects, 20 chapters, 500 sample MCQs for demo.
Test Suite	Unit tests for critical business logic (quiz engine, streak logic, AI calls). Min 70% coverage.
CI/CD Pipeline	GitHub Actions pipeline: lint → test → build → deploy on push to main.
Environment Config	.env.example file with all required environment variables documented.
Design Assets	All Figma design files for UI components, landing page, dashboard.
Deployment Setup	Production environment on Vercel + Railway/DO, fully configured and live.

Feature	Description
Monitoring Setup	Sentry error tracking configured. Basic uptime monitoring (e.g. UptimeRobot).
Training Session	1-hour handover call: how to use admin panel, upload papers, add MCQs, broadcast notifications.

13. Pricing Strategy & Business Model

PrepBuddy's business model is built for long-term dominance — not short-term revenue. The strategy is simple: Launch completely free, build a loyal student base, become the go-to name for MDCAT prep in Pakistan, and then introduce paid tiers once students cannot imagine preparing without PrepBuddy. This is exactly how Duolingo, Notion, and every successful EdTech platform has won their markets.

■ Launch Philosophy: FREE FIRST, MONETISE LATER

When PrepBuddy launches, **100% of all features are completely free** for every student. No credit card, no trial limits, no paywalls. Every MCQ, every past paper, every AI explanation, every streak, every friend challenge — all free. The goal in Phase 1 and 2 is **user acquisition and trust**, not revenue. Once PrepBuddy has 10,000+ active students and strong word-of-mouth, the premium tier activates in Phase 3. Students who joined free are grandfathered with generous free access — they become the biggest advocates of the platform.

13.1 Monetisation Timeline

Phase	Period	Pricing Model	Goal
Phase 1–2 (Launch)	Months 0–6	100% Free — No limits All features open to everyone	Acquire first 10,000 students Build brand & trust
Phase 3 (Growth)	Months 6–12	Freemium introduced Free tier + Premium tier launched	Convert 5–10% to paid Target: 500–1,000 premium users
Phase 4 (Scale)	Months 12–24	Academy / Institutional plans added Annual plans with discount	Target: 5,000+ premium users Break-even on operating costs
Year 2+ (Profit)	Month 24+	Full freemium + B2B Academy model Sponsored content, Academy partnerships	Sustainable profitable business Market leader in MDCAT prep

13.2 Pricing Tiers (Activated in Phase 3 — Month 6+)

When the premium tier goes live, PrepBuddy uses a 3-tier freemium model. The Free tier remains generous enough that students love the platform — but the Premium tier is compelling enough that serious MDCAT aspirants upgrade.

Feature	■ Free (Always Free)	■ Premium PKR 999 / month ≈ PKR 7,999 / year	■ Academy PKR 4,999 / month (Up to 50 students)
Past Paper Access	Last 2 years only	All years (2015–present)	All years + Admin upload
MCQs Available	500 MCQs / subject	Unlimited MCQs	Unlimited + Custom MCQs
Daily Quiz Limit	3 quizzes / day	Unlimited quizzes	Unlimited + Assign to students
AI Explanations (Grog)	5 per day	Unlimited	Unlimited
AI Study Coach Chat	■ Not included	✓ Included	✓ Included
AI Study Plan	■ Not included	✓ Personalised weekly plan	✓ Class-wide plan
Mock Tests	1 per week	Unlimited mocks	Unlimited + Custom mocks
Performance Analytics	Basic (7-day)	Full (all-time) + export	Full + per-student reports
PDF Download	■ Not available	✓ Download all papers	✓ Download + bulk share
Streak Freeze Tokens	1 per month	4 per month	4 per month
Friend Challenges	✓ Unlimited	✓ Unlimited	✓ Unlimited
Group Competition Rooms	Max 3 friends	Up to 10 friends	Up to 50 students
Leaderboard	✓ Visible	✓ + National rank badge	✓ + Academy leaderboard
Question Bookmarks	Up to 50	Unlimited	Unlimited
Concept Summary Cards (AI)	■ Not included	✓ Any chapter	✓ Any chapter
Mnemonic Generator (AI)	■ Not included	✓ Included	✓ Included

Feature	■ Free (Always Free)	■ Premium PKR 999 / month ≈ PKR 7,999 / year	■ Academy PKR 4,999 / month (Up to 50 students)
Exam Readiness Score	■ Not included	✓ Real-time score	✓ Per-student score
Printable Result PDF	■ Not included	✓ Included	✓ Included
Urdu Interface	✓ Included	✓ Included	✓ Included
Offline Mode (PWA)	■ Not included	✓ Included (Phase 4)	✓ Included (Phase 4)
Ads	Minimal, non-intrusive	✓ Ad-free experience	✓ Ad-free
Support	Community forum	Priority email support	Dedicated WhatsApp support

Pricing Notes:

- PKR 999/month for Premium = roughly PKR 33/day — less than one photocopy. Extremely affordable for serious students.
- Annual plan at PKR 7,999 gives students ~33% discount vs monthly — encourages long-term commitment.
- Students who joined during the free launch phase get a **Founding Member** discount: 50% off Premium for life.
- Payment methods: EasyPaisa, JazzCash, Debit/Credit Card via Stripe — all major Pakistani payment methods.
- Academy plan allows coaching academies to manage up to 50 student accounts — ideal for MDCAT prep centres.
- Free trial: 14-day full Premium trial for all new users — no credit card required at signup.
- Student verification discount: Students with a valid CNIC or student ID get an additional 20% off.

13.3 Revenue Projections (Conservative Estimates)

These are realistic, conservative estimates based on standard freemium conversion rates (3–8%) in EdTech platforms in South Asia. These numbers grow significantly with word-of-mouth, social media growth, and academy partnerships.

Milestone	Total Users	Premium Users (5% conversion)	Monthly Revenue (PKR)	Annual Revenue (PKR)
Month 6 (Premium Launch)	10,000	500	499,500	~6,000,000

Milestone	Total Users	Premium Users (5% conversion)	Monthly Revenue (PKR)	Annual Revenue (PKR)
Month 12	35,000	1,750	1,748,250	~21,000,000
Month 18	80,000	4,000	3,996,000	~48,000,000
Month 24 (+ Academy plans)	150,000	7,500 + 50 academies	7,742,500	~93,000,000

13.4 Additional Revenue Streams (Phase 4+)

Feature	Description
Sponsored Content	MDCAT prep books, academies, and medical universities can sponsor newsletter sections or blog posts — clearly labelled as sponsored.
Academy Partnerships	Formal partnerships with top MDCAT academies (Kips, Star, etc.) — PrepBuddy becomes their digital practice platform.
In-App Advertising	Minimal, tasteful ads on the Free tier only — medical book publishers, academy ads. Never intrusive.
Institutional Licensing	Universities and government bodies license PrepBuddy content for their own student portals.
Data Insights (Anonymised)	Sell aggregated, anonymised performance data to PMC or universities to help improve MDCAT question quality.
PrepBuddy Store	Branded merchandise: PrepBuddy notebooks, planners — sold to students who love the brand.

13.5 Why 'Free First' is the Right Strategy for Pakistan

- Pakistani students are extremely price-sensitive. Asking for money at signup kills conversion instantly — even PKR 200 is a barrier.
- Word-of-mouth is the #1 growth channel in Pakistan. A student who loves PrepBuddy tells 10 friends. Free removes all friction to try it.
- Building a large free user base creates a powerful data moat — PrepBuddy knows more about MDCAT prep patterns than anyone else.
- Once students have months of streak history, quiz data, and friends on PrepBuddy, switching to a competitor becomes painful. Retention becomes natural.
- Investors, media, and academies take PrepBuddy more seriously with 50,000 users than with 500 paying users. Free growth builds credibility.

- The MDCAT calendar is fixed (once a year). Free access lets PrepBuddy capture ALL students at the start of their prep cycle — not just those who can pay.

PrepBuddy — Empowering Pakistan's Medical Aspirants, One MCQ at a Time.

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